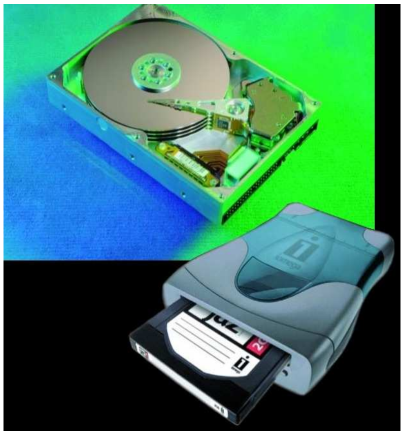
Memory



The memory holds instructions, data that needs to be processed and the results. It is divided into two types, Primary memory is the main memory on the motherboard of the computer, and Secondary memory is the storage media used to store the program or data permanently. The primary memory is of two types : ROM and RAM

Read-Only Memory (ROM)

ROM is a type of memory that normally can only be read, as opposed to RAM which can be both read and written. There are two main reasons that read-only memory is used for certain functions within the PC:

The *values* stored in ROM are always there, whether the power is on or not. A ROM can be removed from the PC, stored for an indefinite period of time, and then replaced, and the data it contains will still be there. For this reason, it is called *non-volatile storage*. A hard disk is also non-volatile, for the same reason, but regular RAM is not.

Read-only memory is most commonly used to store system-level programs that we want to have available to the PC at all times. The most common example is the system BIOS program, which is stored in a ROM called the *system BIOS ROM*. Having this in a permanent ROM means it is available when the power is turned on so that the PC can use it to boot up the system.

The following are the different types of ROMs with a description of their relative modifiability:

- **ROM:** A regular ROM is constructed from ***hard-wired*** logic, encoded in the silicon itself, much the way that a processor is. It is designed to perform a specific function and cannot be changed. This is inflexible and so regular ROMs are only used generally for programs that are static (not changing often) and mass-produced. This product is analogous to a commercial software CD-ROM that you purchase in a store.
- **Programmable ROM (PROM):** This is a type of ROM that can be programmed using special equipment; it can be written to, but only once. This is useful for companies that make their own ROMs from software they write, because when they change their code they can create new PROMs without requiring expensive equipment. This is similar to the way a CD-ROM recorder works by letting you "burn" programs onto blanks once and then letting you read from them many times. In fact, programming a PROM is also called *burning*, just like burning a CD-R, and it is comparable in terms of its flexibility.

- **Erasable Programmable ROM (EPROM):** An *EPROM* is a ROM that can be erased and reprogrammed. A little glass window is installed in the top of the ROM package, through which you can actually see the chip that holds the memory. Ultraviolet light of a specific frequency can be shined through this window for a specified period of time, which will erase the EPROM and allow it to be reprogrammed again. Obviously this is much more useful than a regular PROM, but it does require the erasing light. Continuing the "CD" analogy, this technology is analogous to a reusable CD-RW.
- **Electrically Erasable Programmable ROM (EEPROM):** The next level of eras ability is the *EEPROM*, which can be erased under software control. This is the most flexible type of ROM, and is now commonly used for holding BIOS programs. When you hear reference to a "flash BIOS" or doing a BIOS upgrade by "flashing", this refers to reprogramming the BIOS EEPROM with *a special software program*. Here we are blurring the line a bit between what "read-only" really means, but remember that this rewriting is done maybe once a year or so, compared to real read-write memory (RAM) where rewriting is done often many times per second!

Finally, one other characteristic of ROM, compared to RAM, is that it is much slower, typically having double the access time of RAM or more.

Random Access Memory (RAM)

The kind of memory used for holding programs and data being executed is called *random access memory* or *RAM*. RAM differs from read-only memory (ROM) in that it can be both read and written. It is considered *volatile storage* because unlike ROM, the contents of RAM are lost when the power is turned off. Obviously, RAM needs to be writeable in order for it to do its job of holding programs and data that you are working on. The volatility of RAM also means that you risk losing what you are working on unless you save it frequently.

There are many different types of RAMs, including static RAM (SRAM) and many flavors of dynamic RAM (DRAM).

Static RAM (SRAM)

Static RAM is a type of RAM that holds its data *without external refresh*, for as long as power is supplied to the circuit. This is contrasted to dynamic RAM (DRAM), which *must be refreshed* many times per second in order to hold its data contents. SRAMs don't require external refresh circuitry or other work in order for them to keep their data intact and SRAM is faster than DRAM.

SRAM is superior to DRAM, and we would use it exclusively if only we could do so economically. Unfortunately, 32 MB of SRAM would be prohibitively large and costly, which is why DRAM is used for system memory. SRAMs are used instead for level 1 cache and level 2 cache memory, for which it is perfectly suited; cache memory needs to be very fast, and not very large.

Each SRAM bit is comprised of between four and six transistors, which is why SRAM takes up much more space compared to DRAM, which uses only one (plus a *capacitor*). Because an SRAM chip is comprised of thousands or millions of identical cells, it is much easier to make than a CPU, which is a large die with a non-repetitive structure. This is one reason why RAM chips cost much less than processors do

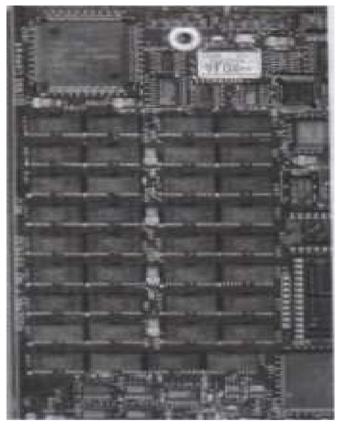


FIGURE RAM soldered onto an ancient motherboard.

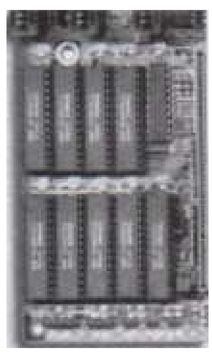


FIGURE SRAM on a motherboard

Dynamic RAM (DRAM)

Dynamic RAM is a type of RAM that only holds its data if it is continuously accessed by special logic called a *refresh circuit*. Many hundreds of times each second, this circuitry reads the contents of each memory cell, whether the memory cell is being used at that time by the computer or not. Due to the way in which the cells are constructed, the reading action itself refreshes the contents of the memory. If this is not done regularly, then the DRAM will lose its contents, even if it continues to have power supplied to it. This refreshing action is why the memory is called *dynamic*.

All PCs use DRAM for their main system memory, instead of SRAM, even though DRAMs are slower than SRAMs and require the overhead of the refresh circuitry. The reason that DRAMs are used is simple: they are much *cheaper* and take up much less space, typically 1/4 the silicon area of SRAMs or less. To build a 64 MB core memory from SRAMs would be very expensive. The overhead of the refresh circuit is tolerated in order to allow the use of large amounts of inexpensive, compact memory. The refresh circuitry itself is almost never a problem; many years of using DRAM has caused the design of these circuits to be all but perfected.

There are many different kinds of specific DRAM technologies and speeds that they are available in. These have evolved over many years of using DRAM for system memory.

Dynamic Random Access Memory (DRAM) used only a single capacitor per bit of data and, thus, cost substantially less than SRAM, but had many disadvantages. The DRAM memory registers (that is, the transistors) required periodic refreshing (every few milliseconds) during which the processor couldn't access the RAM (called a wait state). Refreshing caused DRAM to be slower than SRAM. Additionally, DRAM used more power than SRAM. Cost drives many things, however, so DRAM became the primary system RAM in all computers for many years. DRAM came in two varieties, such as Fast Page Mode (FPM) and Extended Data Out (EDO). FPM was the standard DRAM until edged out by the better technology of EDO.

EDO RAM

Extended Data Out RAM (EDO RAM) is a special type of FPM DRAM that works faster by grabbing larger chunks of data to feed to the CPU. Like regular FPM DRAM, EDO is asynchronous and doesn't use the system clock for timing. EDO RAM is sold in 72-pin SIMMs and 168-pin DIMMs, and it has access times of 50-60 nanoseconds.

SDRAM (Synchronous Dynamic RAM)

Synchronous Dynamic RAM (SDRAM)-the current RAM of choice in most systems-offers a great improvement over FPM and EDO RAM, delivering data in high-speed bursts. Plus, SDRAM runs at the speed of the system bus (thus, synchronously) and has access times in the 8-10 ns range.

Manufacturers rate SDRAM according to the highest speed data bus it runs on, such as 66, 100, and 133 MHz, and name the RAM accordingly: PC66, PC 100, and PC 133. This measurement of speed for SDRAM causes some confusion in older techs because until SDRAM, RAM was measured in nanoseconds. Manufacturers now label SDRAM with both a nanosecond and a megahertz rating to avoid any confusion.

The most common form of SDRAM uses a 168-pin DIMM (Dual Inline Memory Module). SDRAM for laptops comes in two varieties, 72-pin and 144-pin SO (Small Outline) DIMMs.

RDRAM

Rambus Inc. developed Rambus Dynamic RAM (RDRAM). RDRAM uses Rambus channels that have data transfer rates of 800 MHz. If that's not fast enough for you, you can also double the channel width for faster speeds. When the channel rates are doubled, RDRAM can reach data transfer speeds of up to 1.6 GHz. Naturally, RDRAM can only be used in systems that support it.

VRAM (Video RAM)

Video RAM (VRAM) can be found on video cards. VRAM provides faster access than EDO RAM because it can be both read from and written to at the same time, that is, VRAM is dual ported. The speed of VRAM makes it perfect for video.

Physical Memory Organization

SIMMs (Single Inline Memory Module)

Single Inline Memory Modules (SIMMs) were originally created to free up room on the motherboard. These RAM modules are available in either 30-pin or 72-pin layouts. 30-pin SIMMs put out 8 bits of data on the data bus at one time, which makes them 8-bits wide. They come in 1-16MB sticks. 72-pin SIMMs are 32-bits wide and are available in MB to 64MB sticks.

Installing SIMMs is straightforward. You need to insert the SIMM into the memory slot on the motherboard at about a 45-degree angle. After you place the RAM in the correct location, you need to snap it into place, so it's upright or perpendicular to the motherboard. The keyed notches on the SIMMs help to ensure a proper insertion.
To remove SIMMs from the memory slots, you need to push the tab on either side out of the way. After moving the tabs, simply pull the SIMM out of the memory slot.

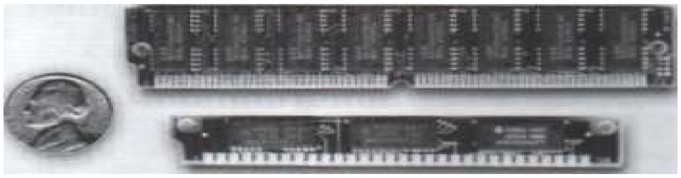


FIGURE 72-pin and 30-pin SIMMs

FIGURE Installing a 72-pin SIMM

DIMMs (Dual Inline Memory Module)

Dual Inline Memory Modules (DIMMs) for regular desktop PCs look similar to SIMMs, but are wider and longer. DIMMs have 168-pins, are 64-bits wide, and range in capacity from 8M B to 256MB sticks. The 144-pin SO DIMMs are also 64-bits wide and come in roughly the same capacities as regular DIMMs. The 72-pin SO DIMMs for laptop PCs, in contrast, are only 32-bits wide.

To install a DIMM into the motherboard, you need to drop it into a blank memory slot and press down firmly. When you've pressed the RAM down into the slot firmly enough, the white locking pins snap into place. DIMMs always have a notch on the pin-edge, so you cannot install them backwards. If a DIMM seems overly resistant to installation, check the orientation of the board!

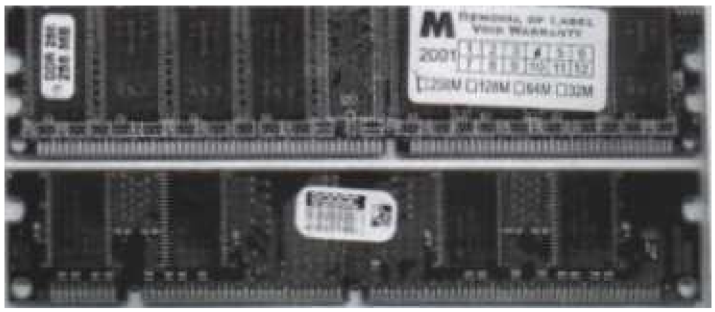


FIGURE 168-pin DIMMs

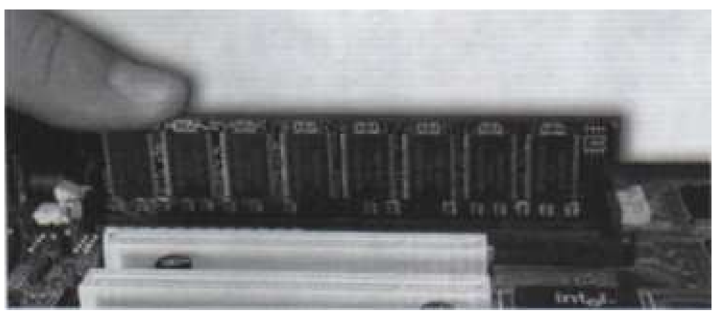


FIGURE Installing a DIMM

RIMM (Rambus Inline Memory Module)

RDRAM comes on Rambus Inline Memory Modules (RIMM). RIMM sticks look like DIMMs, but have 184 pins and are 16-18-bits wide. Rambus memory modules can be purchased in 32, 48, 64, 96, 128, and 256MB sizes, and have 600, 700, and 800 MHz data transfer rates.
Installing RIMMs is a straightforward affair, similar to installing a DIMM, but you have to jump through a hoop or two. Some motherboards require you to place the RIMMs in every other slot, for example, if you install only two RIMMs. Further, each unpopulated slot must have a special CRIMM stick installed. A CRIMM is not RAM, but rather a terminator for open slots.

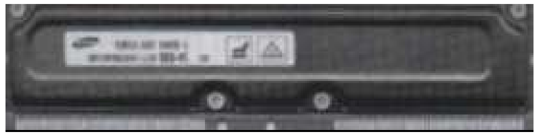


FIGURE RIMM

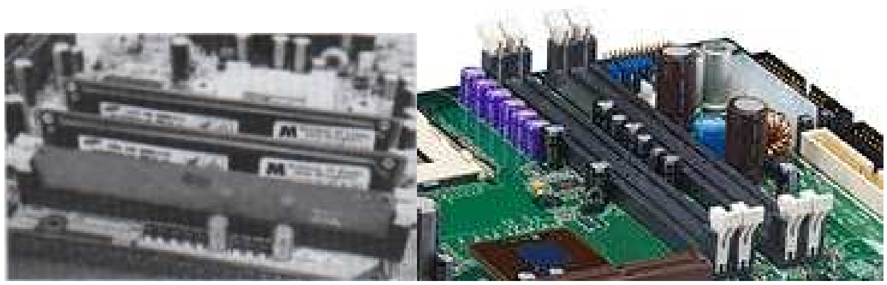


FIGURE CRIMM and RIMM

Memory Banks

Most memory devices are installed in sets (or banks). The amount of memory in the bank can vary, depending on how much you wish to add, but there must always be enough data bits in the bank to fill each bit position. For example, the 8086 is a 16-bit microprocessor (two bytes). This means that two extra bits are required for parity, providing a total of 18 bits. Thus, one bank is 18 bits wide. You might fill the bank by adding eighteen 1-bit DIPs or two 30-pin SIMMs. As another example, an 80486DX is a 32-bit CPU, so 36 bits are needed to fill a bank (32 bits plus 4 parity bits). If you use 30-pin SIMMs, you will need four to fill a bank. If you use 72-pin SIMMs, only one is needed. Notice that the size of the memory in MB does not really matter, so long as the entire bank is filled.

There is more to filling a memory bank than just installing the right number of bits. Memory amount, memory matching, and bank order are three additional considerations. First, you must use the proper memory amount that will bring you to the expected volume of total memory. Suppose a ***Pentium*** system has 8MB already installed in Bank 0, and you need to put another 8MB into the system in Bank 1.

Another bank requirement demands memory matching - using SIMMs/DIMMs of the same size and speed within a bank. For example, when adding multiple SIMMs to a bank, each SIMM must be rated for the same access speed and share the same memory configuration.

You should also be sure to use the same *type* of memory (i.e., EDO, FPM, SDRAM, etc.). For example, if your motherboard is designed to use EDO RAM, and you have EDO RAM already installed, you should be sure to install more EDO RAM. Some motherboard designs allow you to mix memory types, but mixing memory types on other (especially older) motherboards might cause the system to malfunction.

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